

RESEARCH PAPER: QUANTUM ALGORITHMS

Abstract

This paper presents a novel approach to quantum error correction protocols that achieves a 15% reduction in physical qubit overhead.

1. Introduction

Quantum computing holds immense promise for cryptography and material science. However, error rates in NISQ devices remain a challenge. Our proposed protocol, Q-Shield, addresses surface code inefficiencies.

2. Methodology

We simulated 10,000 circuits using Qiskit.
Parameters: $T_1=50\mu s$, $T_2=50\mu s$, 2-qubit gate error 0.1%.

3. Results

The Q-Shield protocol outperformed standard Steane codes in all regimes. Threshold improved from 1.2% to 1.45%.